



Parametric Modelling of Diffusion MRI with Uncertainty Estimation (Report)

Computational Modelling for Biomedical Imaging

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1 Parameter Estimation and Mapping

1.1 Ball-and-Stick Model Fitting

The ball-and-stick model describes the diffusion-weighted MRI signal as arising from two compartments: water restricted to axons (intra-cellular), modelled by a stick, and freely diffusing extra-cellular water, modelled by an isotropic ball. For a measurement with b-value b and gradient direction $\hat{\mathbf{q}}$, the signal is $S(b, \hat{\mathbf{q}}) = S_0[f e^{-bd(\hat{\mathbf{q}} \cdot \mathbf{n})^2} + (1 - f)e^{-bd}]$, where S_0 is the baseline signal, d the diffusivity (mm^2/s), $f \in [0, 1]$ the intra-cellular volume fraction, and $\mathbf{n} = [\cos \phi \sin \theta, \sin \phi \sin \theta, \cos \theta]^\top$ the unit fibre direction parameterised by polar angles (θ, ϕ) . The vector $\mathbf{x} = (S_0, d, f, \theta, \phi)$ was estimated by minimising the sum of squared differences (SSD) objective function $\mathcal{L}(\mathbf{x}) = \sum_{k=1}^K (A_k - S(b_k, \hat{\mathbf{q}}_k; \mathbf{x}))^2$ over $K = 108$ measurements for voxel [92, 65, 72], using `fminunc` with starting point $\mathbf{x}_0 = (3.5, 3 \times 10^{-3}, 0.25, 0, 0)$.

The fitted parameter values of the unconstrained fit can be found in Table 1, with $\text{RESNORM} = 2.874 \times 10^7$. The parameters d and f are non-physical: d is negative and $f > 1$. As $d \rightarrow 0^-$, both exponentials in the ball-and-stick model collapse to ≈ 1 , making the model insensitive to diffusion weighting or gradient direction. This is visible in Figure 1: the model predicts a near-flat signal at approximately 3500 a.u., systematically underestimating the high-signal measurements.

The expected SSD under i.i.d. Gaussian noise with standard deviation $\sigma = 200$ is $\mathbb{E}[\mathcal{L}] = K\sigma^2 = 4.32 \times 10^6$. The obtained RESNORM is approximately $6.7\times$ larger, strongly indicating that no global minimum was found. Note that diffusion MRI noise actually follows a Rician rather than Gaussian distribution, and the ball-and-stick model is itself an approximation, so the SSD benchmark should be interpreted as an order-of-magnitude reference rather than a precise threshold [1].

1.2 Parameter Constraints via Transformation

To enforce physically valid parameters, the transformation method [1] was applied: $S_0 = \alpha_1^2$, $d = \alpha_2^2$, and $f = \sin^2(\alpha_3)$, ensuring $S_0 > 0$, $d > 0$, and $f \in [0, 1]$ for all real α . The starting point was transformed accordingly as $\alpha_{1,0} = \sqrt{S_{0,0}}$, $\alpha_{2,0} = \sqrt{d_0}$, and $\alpha_{3,0} = \arcsin(\sqrt{f_0})$. The angular parameters θ and ϕ require no constraint since $\mathbf{n}(\theta, \phi)$ automatically satisfies $\|\mathbf{n}\| = 1$. Alternative mappings (e.g. sigmoid or exponential) could also restrict $f \in [0, 1]$, but empirical tests showed that only $\sin^2$ and sigmoid yielded the lowest RESNORM; $\sin^2$ was therefore adopted.

The constrained fit ensured valid estimates for S_0 , d , and f , reducing RESNORM to 5.872×10^6 , only $1.36\times$ above the expected SSD. Compared to the unconstrained fit (Fig. 1), the constrained model (Fig. 2) better matches the high-signal measurements, reflecting a larger S_0 and improved overall fit between model and data. The parameter values can be found in Table 2.

1.3 Global Minimum Analysis

Because gradient-based optimisers converge to the nearest local minimum [1], the constrained fit was repeated $N_{\text{runs}} = 1000$ times from randomly perturbed starting points [2]. Perturbations were generated as $\mathbf{p} = \mathbf{x}_0 + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\boldsymbol{\sigma} = (10^3, 10^{-3}, 0.1, 0.5, 1.0)$, where the scales were chosen to reflect the typical magnitude of each parameter.

The RESNORM histogram (Fig. 3) shows two peaks and an outlier around 3.5×10^8 , indicating convergence to different minima. The smallest RESNORM occurred in 61% of runs and was taken as the global minimum candidate. [1] notes that the more runs that converge to the same lowest value, the higher the confidence this is a global minimum. The required number of runs to find this minimum in at least one run with probability $\geq 95\%$ is $N_{95} = \lceil \log(0.05) / \log(1 - p_{\text{global}}) \rceil = 4$ [2].

The same procedure was applied to additional voxels (Fig. 4), with results summarised in Table 3. The grey matter voxel shows the highest p_{global} , indicating a more stable fit and fewer runs required in this voxel to find a global

minimum. The CSF voxel has the lowest p_{global} and highest N_{95} , reflecting the weaker directional signal and poorer model fit due to the largely isotropic diffusion in CSF.

1.4 Parameter Mapping

Parameter mapping was performed over the middle axial slice ($z = 72$) by fitting the ball-and-stick model in every voxel within a brain mask. The mask was obtained by thresholding the mean $b = 0$ image to exclude background voxels (Figure 5). Each masked voxel was fitted using $N_{\text{runs}} = 4$ random-start constrained fits, retaining the result with the lowest RESNORM. The resulting maps of S_0 , d , f , and RESNORM are shown in Figure 6. Higher stick fractions f appear in white matter tracts, indicating strongly oriented diffusion, whereas lower values in CSF reflect more isotropic diffusion. A quiver plot of the fibre direction $\mathbf{n}$ weighted by the volume fraction f is shown in Figure 7. To improve readability, the vector field was spatially subsampled by plotting one arrow per 3×3 voxel block. White matter regions have higher f values and aligned fibre directions, reflecting major fibre bundles.

1.5 Improving Optimisation

Two strategies were investigated to improve the reliability and efficiency of the optimisation: explicit bounds via `fmincon`, and DT-informed starting points.

Explicit bounds via `fmincon`. Constraints were imposed directly with lower bounds $\mathbf{l} = (10^{-6}, 10^{-6}, 0, -\infty, -\infty)$ and upper bounds $\mathbf{u} = (\infty, \infty, 1, \infty, \infty)$. Although the parameter space is correctly constrained, the interior-point solver is $26\times$ slower per run than the transformation method. Combined with a lower $p_{\text{global}} = 0.51$, this raises N_{95} and the total slice mapping time to 66.4 min, making it impractical for whole-brain mapping.

Informed starting point from the diffusion tensor (DT). The DT model, $\log S_k \approx \log S_0 - b_k \hat{\mathbf{q}}_k^\top \mathbf{D} \hat{\mathbf{q}}_k$, was fitted via weighted least squares [1], and its principal eigenvector $\mathbf{e}_1$, mean diffusivity MD, and fractional anisotropy FA were used to initialise $(\mathbf{n}, d, f)$. Among five tested mappings (Table 4), strategy M2 (MD + FA + $\mathbf{e}_1$) performed best, achieving $p_{\text{global}} = 0.793$ and $N_{95} = 2$ at no additional per-run cost. The estimated slice time of 4.2 min is slightly higher than the transformation method alone due to the additional DT fitting step, but the substantially improved reliability justifies this overhead. The information is summarised in Table 5.

2 Uncertainty Estimation

2.1 Bootstrap Estimation

The classical bootstrap generates an empirical distribution of parameter estimates that approximates the posterior distribution $p(\mathbf{x} \mid A)$ without requiring assumptions about the underlying noise model, as illustrated in Figure 8 [1]. Figure 8 compares the 2σ interval, which corresponds to approximately 95% coverage under a Gaussian assumption, with the empirical 95% interval obtained directly from the bootstrap samples. The closer these two ranges are, the more closely the parameter distribution follows a Gaussian shape. Following the recommendation of [3], 1000 bootstrap samples were used. Figure 9 shows the convergence of the bootstrap statistics, indicating that the chosen number of bootstrap iterations is sufficient, as the estimates stabilise for all parameters [1].

Figure 10 shows the same plot for other voxels tested, the same voxels as indicated in Figure 4. In general, S_0 appears approximately Gaussian distributed, with the 2σ interval closely matching the empirical 95% interval for all voxels tested. For d and f , a larger discrepancy between the two intervals is observed for the white matter and CSF voxels, whereas the grey matter voxel shows relatively stable behaviour across all parameters. A detailed overview of the confidence intervals across the different voxels and the different parameters is listed in Table 6-8.

2.2 MCMC (Metropolis-Hastings)

Uncertainty was additionally estimated using MCMC. The Markov chain was run for 100000 iterations, of which the first 10000 iterations were discarded as burn-in, as indicated in Figure 12. After burn-in and thinning, 18000 effective samples were retained for posterior analysis. Figure 11 shows that the running mean and running standard deviation

of S_0 , d , and f stabilise after approximately 4000 retained samples, while the trace plots in Figure 13 confirm that the chain explores the parameter space without getting stuck in a particular region for many iterations.

The proposal widths were tuned following the strategy noted in [1] to obtain an overall acceptance rate in the recommended range of 20-50%. Each proposal width was tuned sequentially by increasing its step size until the corresponding acceptance rate was within the recommended 20-50% range. After tuning the parameters individually, all proposal widths were jointly rescaled by a common factor to obtain a suitable overall acceptance rate. The final proposal widths are $\sigma_{S_0} = 38.12$, $\sigma_d = 1.906 \times 10^{-5}$, $\sigma_f = 0.01144$, $\sigma_\theta = 0.04575$, and $\sigma_\phi = 0.05719$, yielding an acceptance rate of 26.5%. The posterior distributions in Figure 14 are close to Gaussian, with the empirical 2σ intervals closely matching the Gaussian fits for S_0 , d , and f . Figure 15 and Table 9 show that the MCMC and bootstrap estimates largely overlap, with very similar posterior means and only slightly narrower uncertainty intervals for MCMC.

3 Model Selection

3.1 Model Fitting to ISBI data

Figure 16 shows that the ISBI 2015 dataset contains many low b-value measurements and a clear attenuation of the normalised signal with increasing b-value. A DT-informed starting point was used, since the normalised data are on a very different scale from the raw data in Part 1. Following [2], the stick fraction was constrained using a cosine-based encoding and the ball-and-stick model was fitted with a Levenberg-Marquardt optimiser [2, 4]. The DT-based initialisation is reported in Table 10, while the best-fit parameters are given in Table 11. The same minimum was recovered in all 1000 runs, giving $p_{\text{global}} = 1$, $N_{95} = 1$, and a minimum RESNORM of 15.106. This is $2.61\times$ above the expected RESNORM of 5.7792, suggesting that the ball-and-stick model does not fully explain the data and another model may be better.

Figures 17 and 18 support this conclusion. In Figure 17, the fitted signal follows the overall decay trend, but the residuals show several clear outliers and systematic misfit beyond the expected $\pm 2\sigma$ range. Figure 18 shows strong angular dependence: when the gradient is parallel to the fibre direction, that is $|\cos(\alpha)| \approx 1$, the signal is much lower for high b-values, indicating stronger attenuation along the fibre direction. For b-values close to zero, the signal remains close to one across all directions.

3.2 Other Models

Four models were fitted to the ISBI 2015 data using Levenberg-Marquardt optimisation with parameter constraints enforced via the transformation method. Each compartment model was fitted from 1000 randomly perturbed starting points, initialised hierarchically: the DT (linear least squares) informed the ball-and-stick, which in turn informed both zeppelin variants. Perturbations were applied in physical parameter space with scales reflecting each parameter's plausible range, following the strategy in Part 1. Table 12 and Figure 19-20 summarise the results. The single-compartment DT fits poorly, as it cannot capture the multi-exponential signal decay visible at high b-values. All two-compartment models substantially improve the fit. The zeppelin-and-stick achieves the lowest RESNORM (10.82; ratio 1.87), followed by the zeppelin-and-stick with tortuosity (11.61; 2.01). The improvement from ball to zeppelin suggests that the main limitation of the ball-and-stick model is its assumption of isotropic extracellular diffusion. Allowing this compartment to be anisotropic gives a better description of the measured signal.

3.3 Ranking Models using AIC and BIC

Figure 21 and Table 13 shows that both AIC and BIC give a consistent ranking, with the zeppelin-and-stick model clearly preferred, achieving the best balance between goodness of fit and model complexity. The diffusion tensor performs worst by a large margin, showing that a single-compartment Gaussian model is not sufficient for this multi-shell dataset. Among the two-compartment models, allowing anisotropic extracellular diffusion improves the fit, which explains why both zeppelin-based models outperform ball-and-stick.

Bibliography

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